Machine Learning

AIML CZG565

M10 : Unsupervised Learning

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Contact Session 14

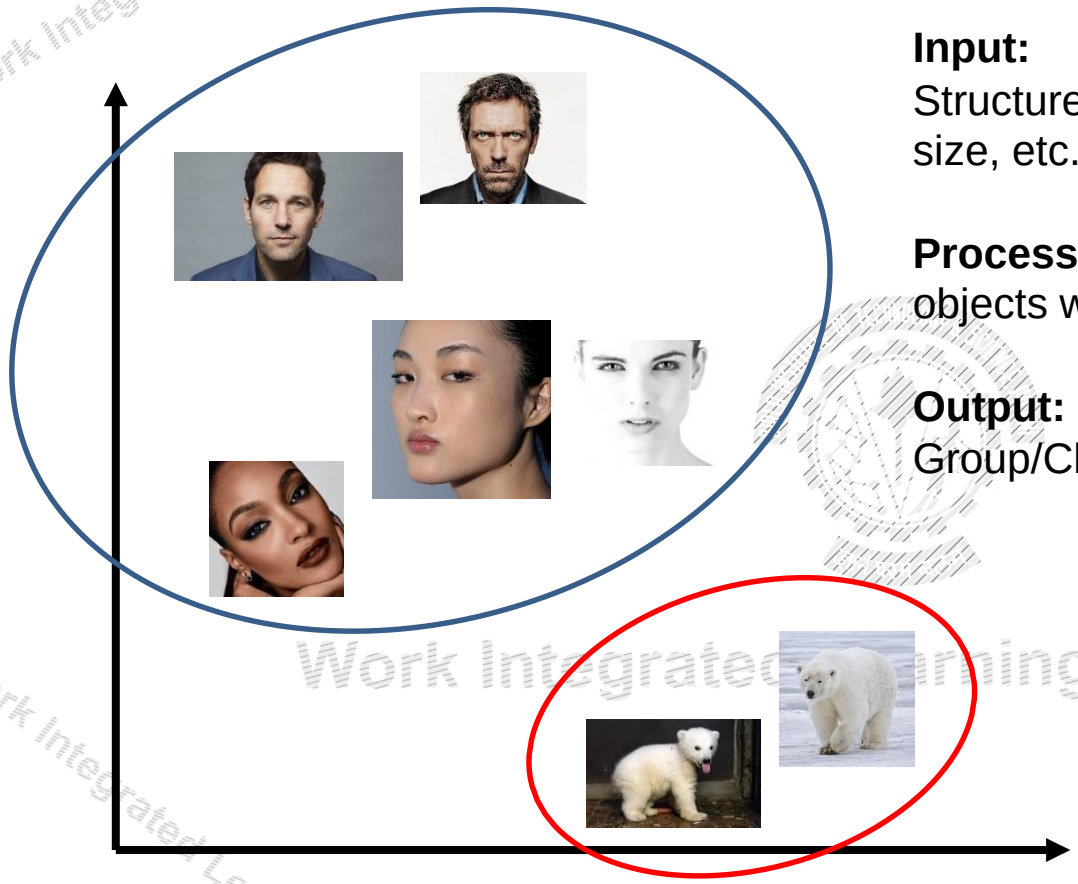
Module –Unsupervised Learning

- Mixture Models
- Expectation Maximization (EM) Algorithm
- K-means Clustering

Work Integrated Learning Programmes

Unsupervised discovery

Task : Group the objects based on the characterisitics features



Input:
Structured data of features like colour , shape, size, etc.,

Process: Find the similarity/dissimlarity between objects with features as the defining dimensions

Output:
Group/Cluster of objects

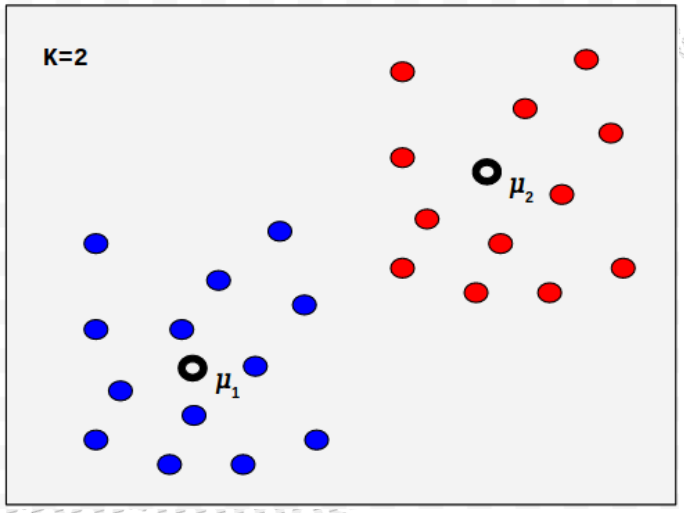
K-Means

Work Integrated Learning Programmes

K-Means Algorithm

- Works iteratively to find $\{\mu_k\}$and $\{r_{nk}\}$ such that J is minimized

$$J = \sum_{n=1}^{N} \sum_{k=1}^{K} r_{nk} \|\mathbf{x}_n - \boldsymbol{\mu}_k\|^2$$



- Iteration involves two key steps
  - Find $\{r_{nk}\}$, fixing $\{\mu_k\}$ to minimize J
  - Find $\{\mu_k\}$, fixing $\{r_{nk}\}$ to minimize J
- Let us look at each of these steps

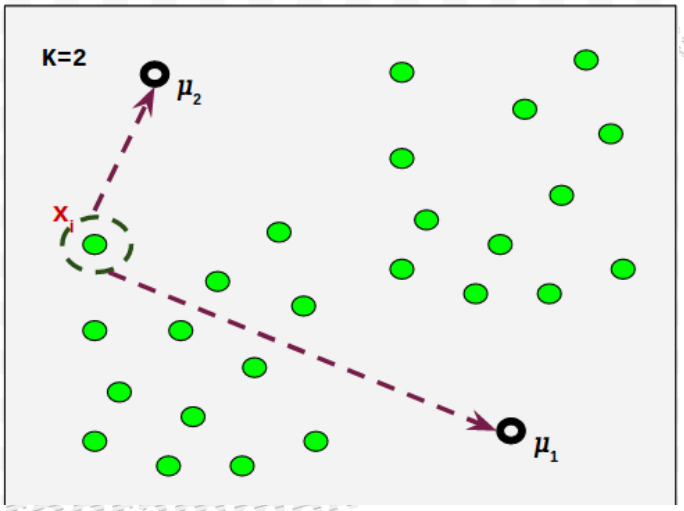
$r_{nk} \in \{0, 1\}$, where $k = 1, \ldots, K$

Binary Indicator variable that indicates point xn belongs to which cluster

$$r_{nk} = \begin{cases} 1 & \text{if } k = \arg\min_j \|\mathbf{x}_n - \boldsymbol{\mu}_j\|^2 \\ 0 & \text{otherwise.} \end{cases}$$

K-Means Algorithm

A sample E-Step



K-Means Algorithm

A sample E-Step

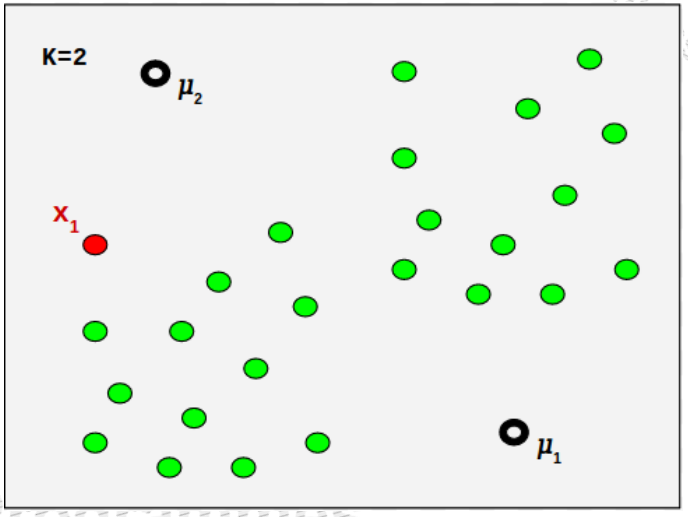
Expectation Step

Initialize random clusters

Two - Phases:

1. Assign / Re-assign data points to clusters based on minimum distance to cluster centers
1. Re-compute the cluster means

Repeat steps 1-2 till convergence

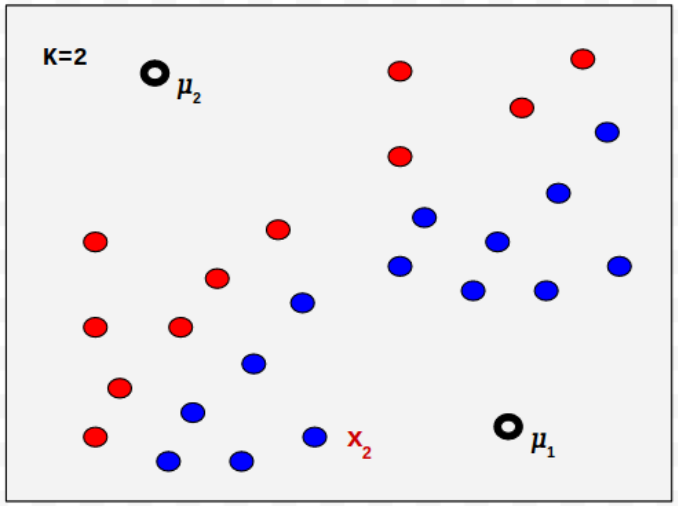


K-Means Algorithm

E-Step:

For all $x_t \in X$:

$$r_{nk} = \begin{cases} 1 & \text{if } k = \arg\min_j \|\mathbf{x}_n - \boldsymbol{\mu}_j\|^2 \\ 0 & \text{otherwise.} \end{cases}$$



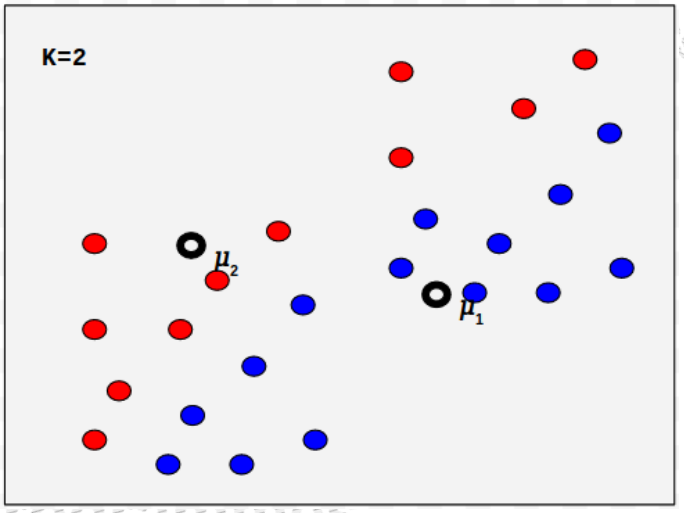
E-step: each data point is assigned to one of the two clusters

K-Means Algorithm

<u>M-Step:</u>

For all μ_k[where k = 1,2,...,K] :

$$\boldsymbol{\mu}_k = \frac{\sum_n r_{nk}\mathbf{x}_n}{\sum_n r_{nk}}$$



re-computign the cluster center as mean of the points assigned to the corresponding cluster

K-Means Algorithm

Algorithm:

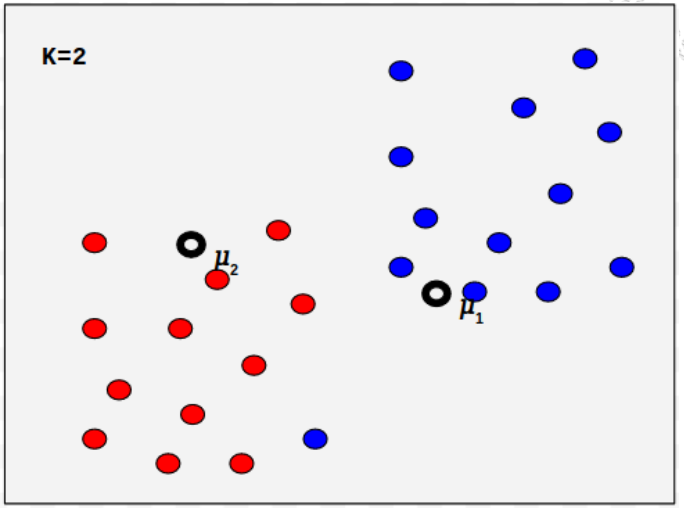
_____Initialize μ_k[where k = 1,2,...,K]

Repeat

E-Step [as defined earlier]

M-Step [as defined earlier]

Until convergence of μ_k .



E-Step in the second iteration

K-Means Algorithm

Algorithm:

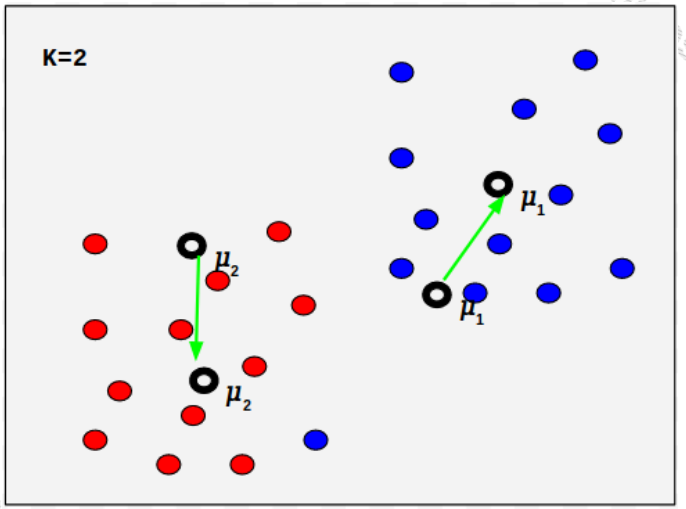
_____Initialize μ_k[where k = 1,2,...,K]

Repeat

E-Step [as defined earlier]

M-Step [as defined earlier]

Until convergence of μ_k .



M-Step in the second iteration

K-Means Algorithm

Algorithm:

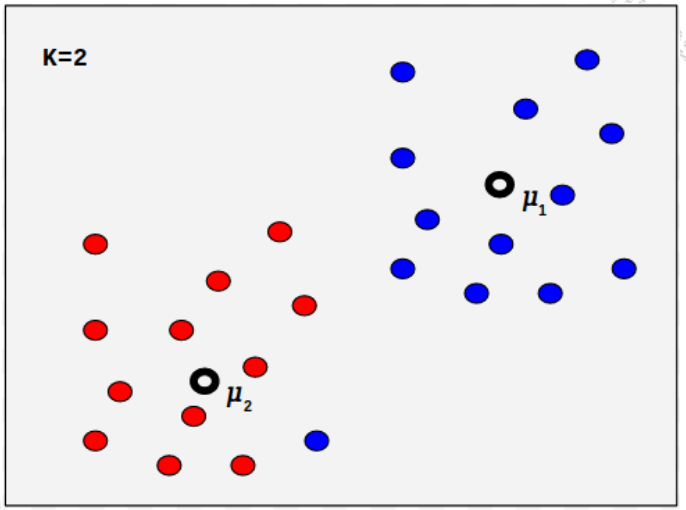
_____Initialize μ_k[where k = 1,2,...,K]

Repeat

E-Step [as defined earlier]

M-Step [as defined earlier]

Until convergence of μ_k .



M-Step in the second iteration

Example

Consider the analysis of weights of individuals and their respective blood glucose levels as given below:

a) Identify the clusters using K-means clustering (k=2) for the given data, assuming candidate 1 and 2 as initial centroids.

b) How many iterations does it take before termination?

Candidate	Weight	Glucose level
1	72	185
2	56	170
3	60	168
4	68	179
5	72	182
6	77	188
7	70	180
8	84	183

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Example

Candidate	Cluster1(72, 185)	Cluster2(56, 170)
3(60, 168)	$=17^2 +12^2$	$= 2^2+4^2$
4(68 179)	$=6^2+4^2$	$= 9^2+12^2$
5(72, 182)	$= 3^2+0^2$	$= 12^2+16^2$
6(77, 188)	$= 3^2+5^2$	$= 18^2+11^2$
7(70, 180)	$= 5^2+2^2$	$= 10^2+14^2$
8(183, 84)	$= 2^2+12^2$	$= 13^2+28^2$

After 1st iteration, the cluster groups are C1{1, 4, 5, 6, 7, 8} and C2{2, 3}

Re-computing centroids:

C1([72+68+72+77+70+84]/6, [185 +179+182+188+180+183]/6) =C1(73.83, 182.83,)

C2([56 +60]/2, [170+168]/2) =C2(58, 169)

2nd Iteration –computing distances from resulting centroids:

Candidate	Cluster1(182.83, 73.83)	Cluster2(169, 58)
1(185, 72)	$(182.83 -185)^2 +(73.83 -72)^2$ $=2.17^2 +1.83^2$	$(169 -185)^2 +(58 -72)^2$ $= 16^2+14^2$
2(170, 56)	$(182.83 -170)^2 +(73.83 -56)^2$ $=12.83^2 +17.83^2$	$(169 -170)^2 +(58 -56)^2$ $= 1^2+2^2$
3(168, 60)	$(182.83 -168)^2 +(73.83 -60)^2$ $=14.83^2 +13.83^2$	$(169 -168)^2 +(58 -60)^2$ $= 1^2+2^2$
4(179, 68)	$(182.83 -179)^2 +(73.83 -68)^2$ $=3.83^2 +5.83^2$	$(169 -179)^2 +(58 -68)^2$ $= 10^2+10^2$
5(182, 72)	$(182.83 -182)^2 +(73.83 -72)^2$ $=0.83^2 +1.83^2$	$(169 -182)^2 +(58 -72)^2$ $= 13^2+14^2$
6(188, 77)	$(182.83 -188)^2 +(73.83 -77)^2$ $=5.17^2 +3.17^2$	$(169 -188)^2 +(58 -77)^2$ $= 19^2+19^2$
7(180, 70)	$(182.83 -180)^2 +(73.83 -70)^2$ $=2.17^2 +1.83^2$	$(169 -180)^2 +(58 -70)^2$ $= 11^2+12^2$
8(183, 84)	$(182.83 -183)^2 +(73.83 -84)^2$ $= 0.17^2+10.17^2$	$(169 -183)^2 +(58 -84)^2$ $= 14^2+26^2$

After 2nd iteration, the cluster groups are C1{1, 4, 5, 6, 7, 8} and C2{2, 3}; there is no change in cluster groups hence the algorithm terminates.

Evaluating K-means Clusters

- Most common measure is Sum of Squared Error (SSE)
  - For each point, the error is the distance to the nearest cluster
  - To get SSE, we square these errors and sum them.

$$SSE = \sum_{i=1}^{K} \sum_{x \in C_i} dist^2(m_i, x)$$

  - x is a data point in cluster C_i and m_i is the representative point for cluster C_i
    - can show that m_i corresponds to the center (mean) of the cluster
  - Given two clusters, we can choose the one with the smallest error
  - One easy way to reduce SSE is to increase K, the number of clusters
    - A good clustering with smaller K can have a lower SSE than a poor clustering with higher K

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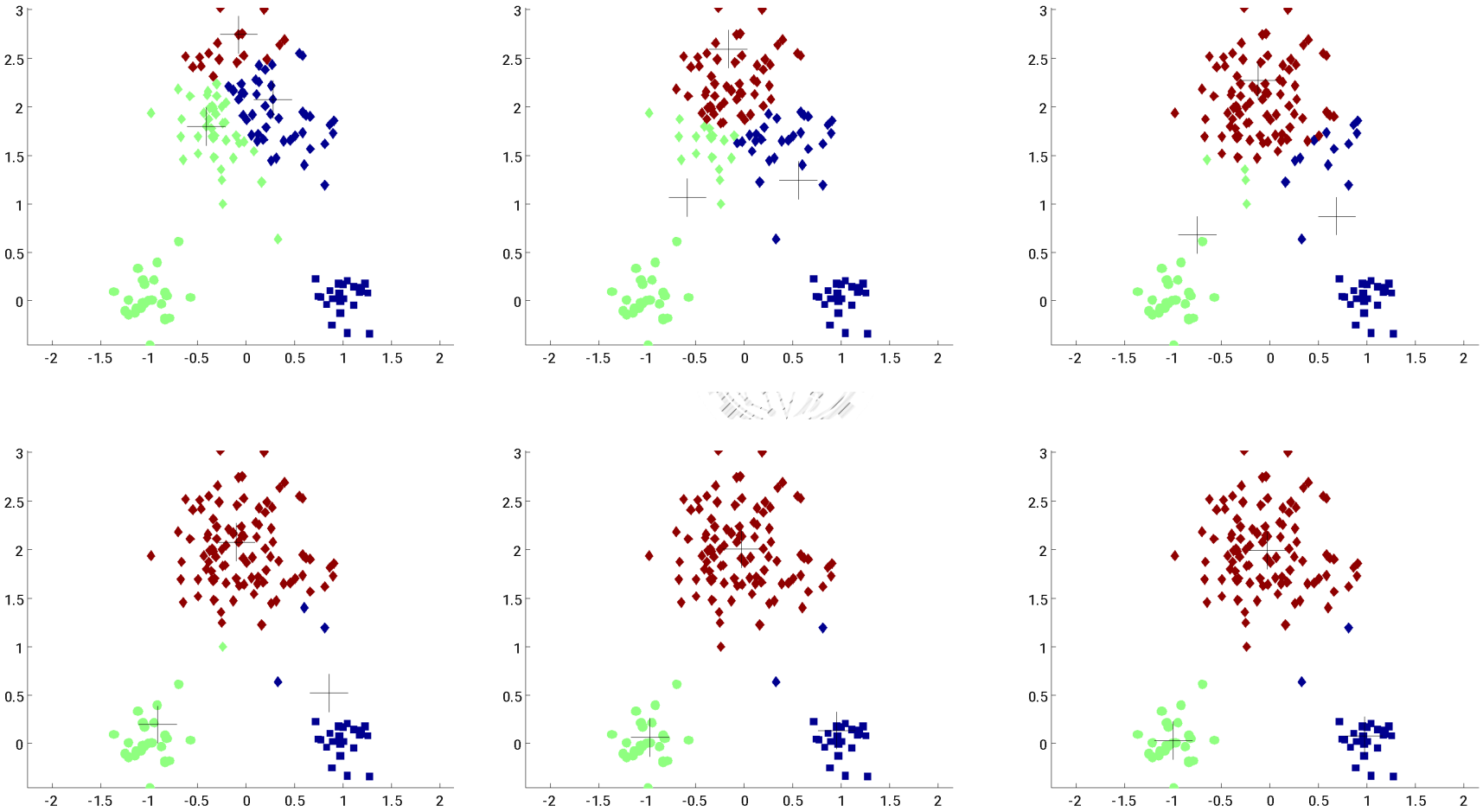
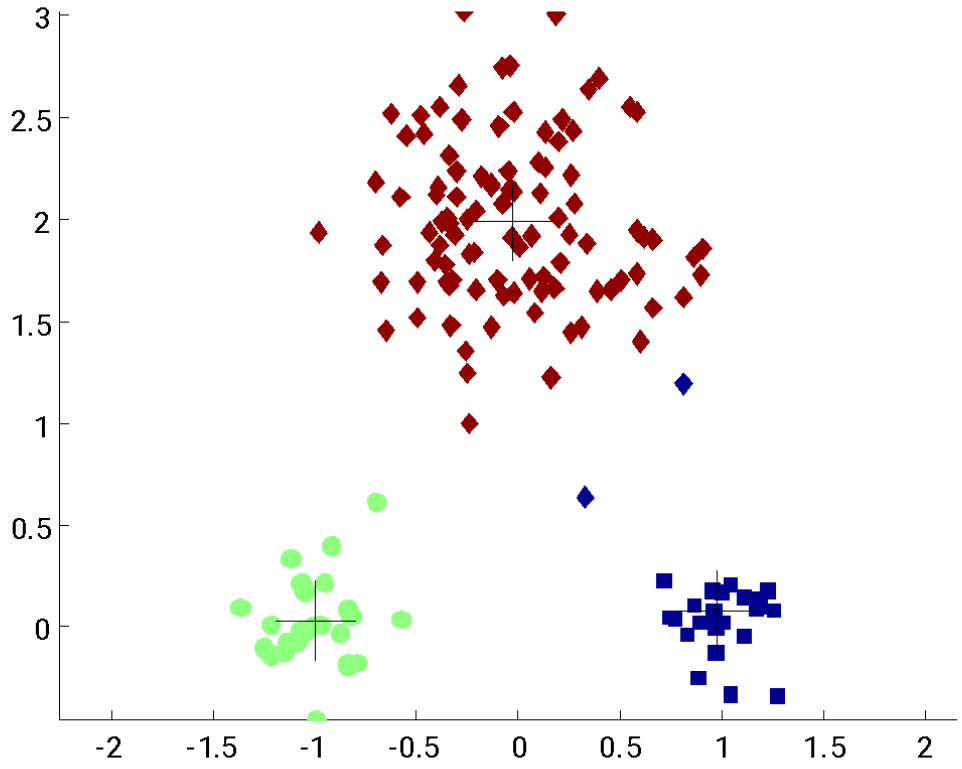
Importance of Choosing Initial Centroids



Importance of Choosing Initial Centroids



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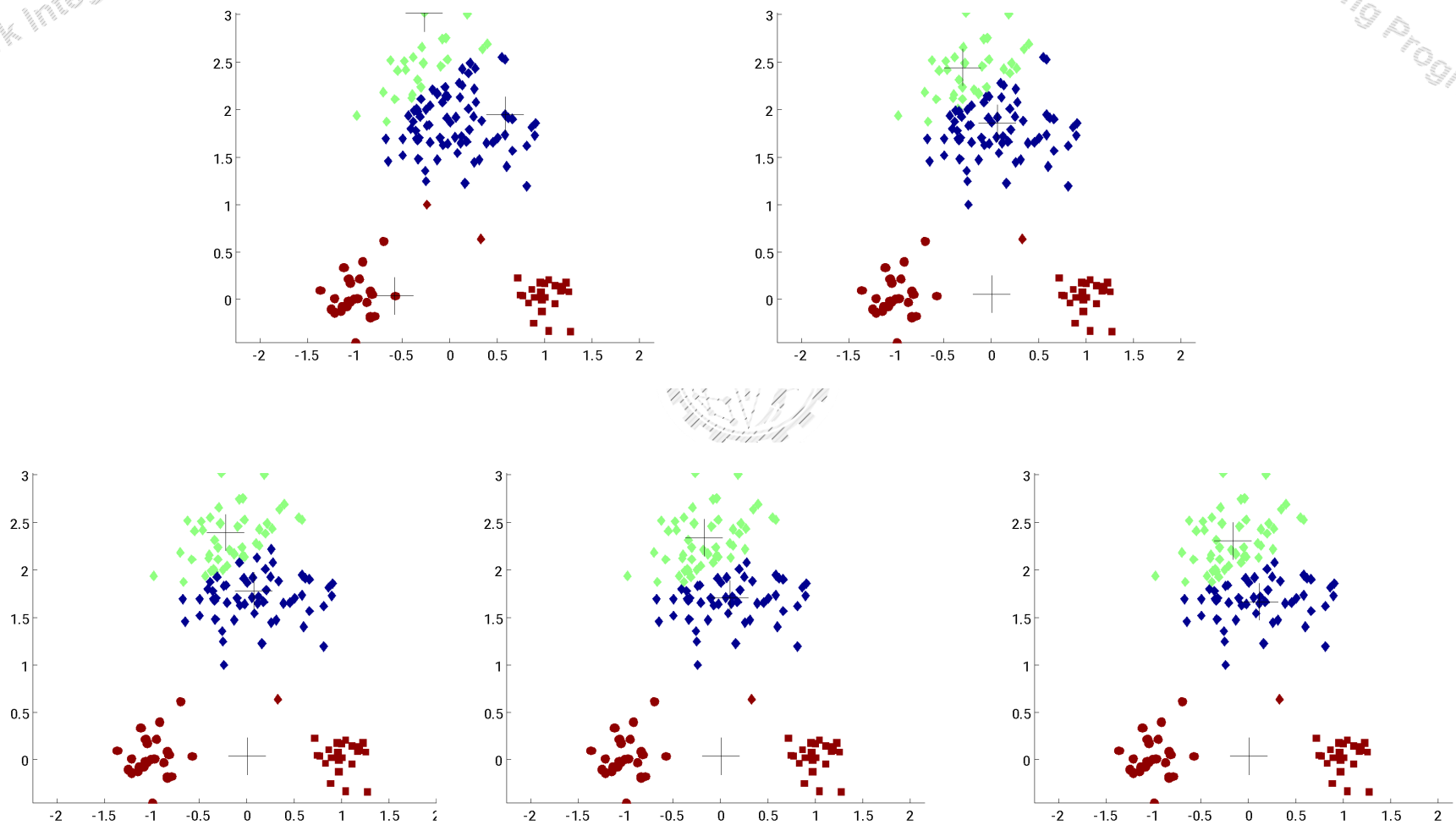
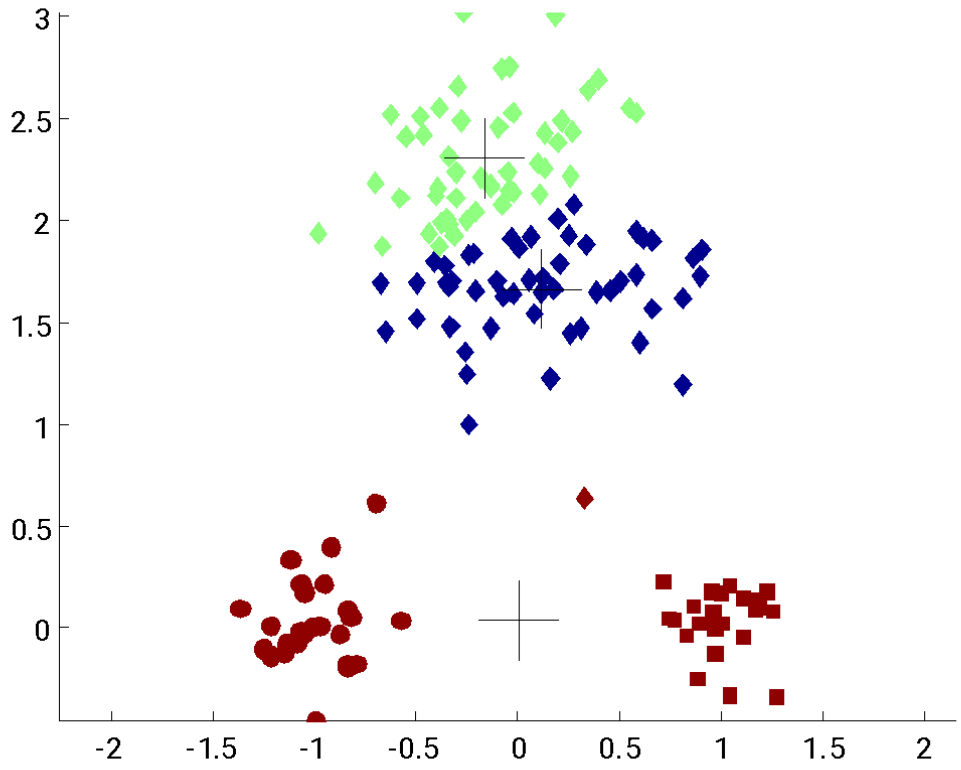
Importance of Choosing Initial Centroids …



Importance of Choosing Initial Centroids …



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Limitations of K-means

- K-means has problems when clusters are of differing
  - Sizes
  - Densities
  - Non-globular shapes

- K-means has problems when the data contains outliers.

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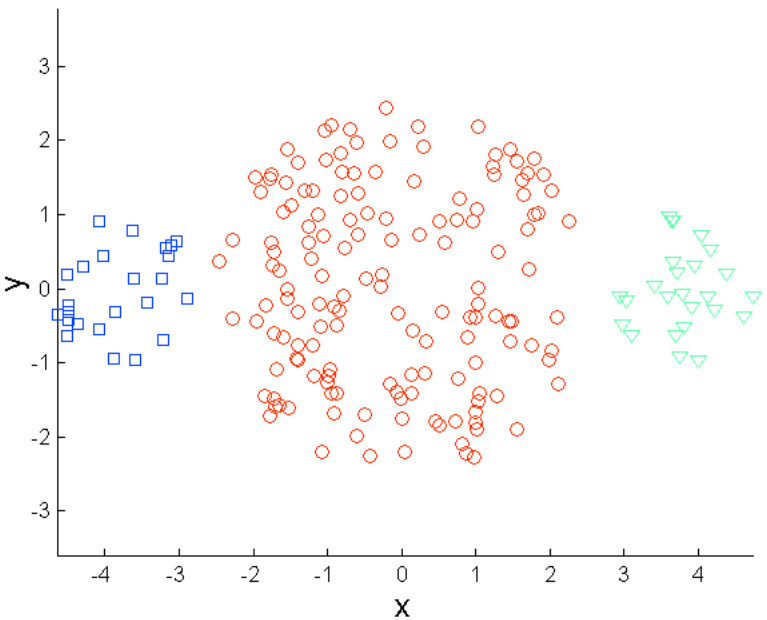
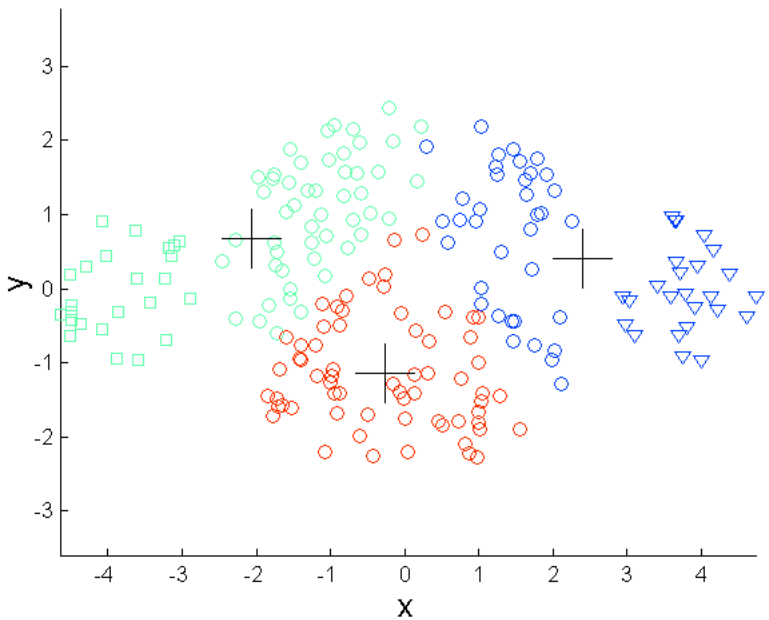
Limitations of K-means: Differing Sizes



Original Points



K-means (3 Clusters)

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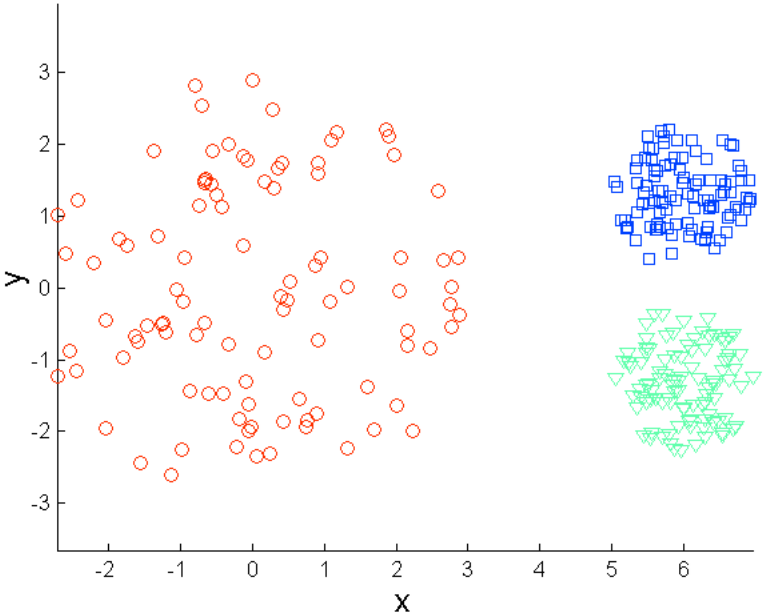
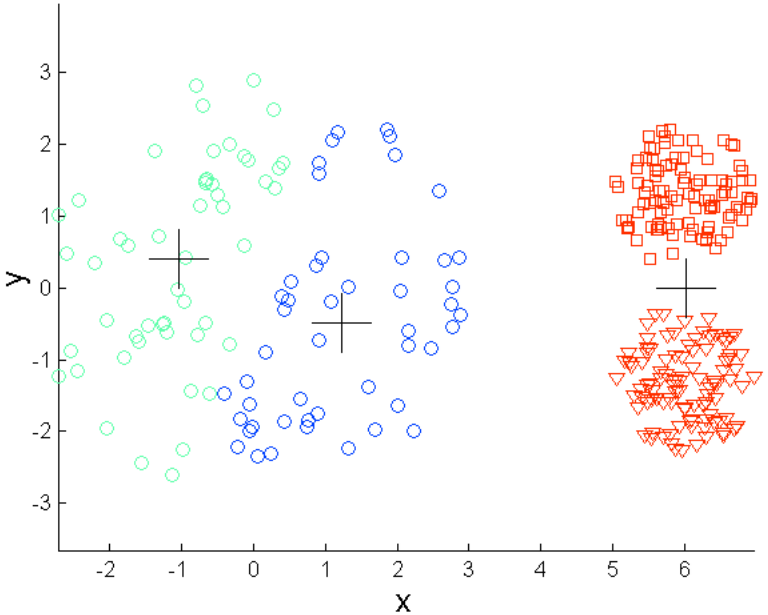
Limitations of K-means: Differing Density



Original Points



K-means (3 Clusters)

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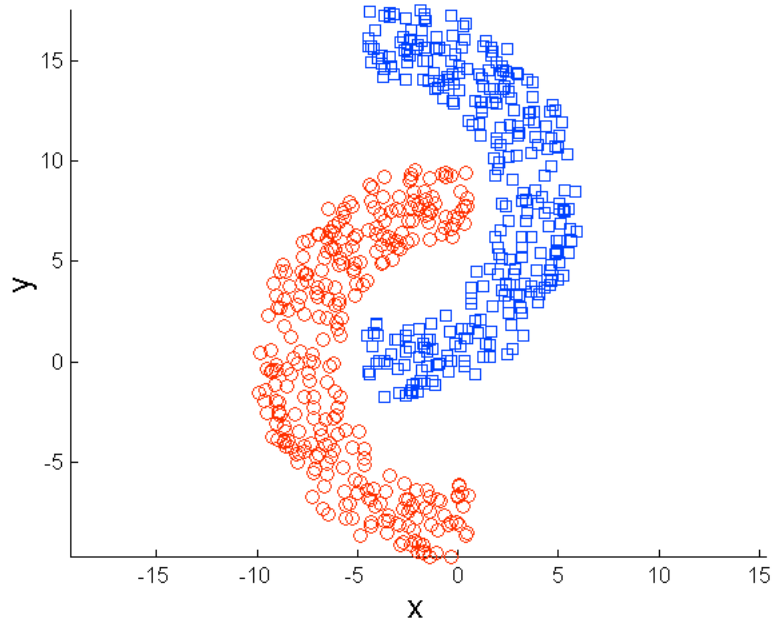
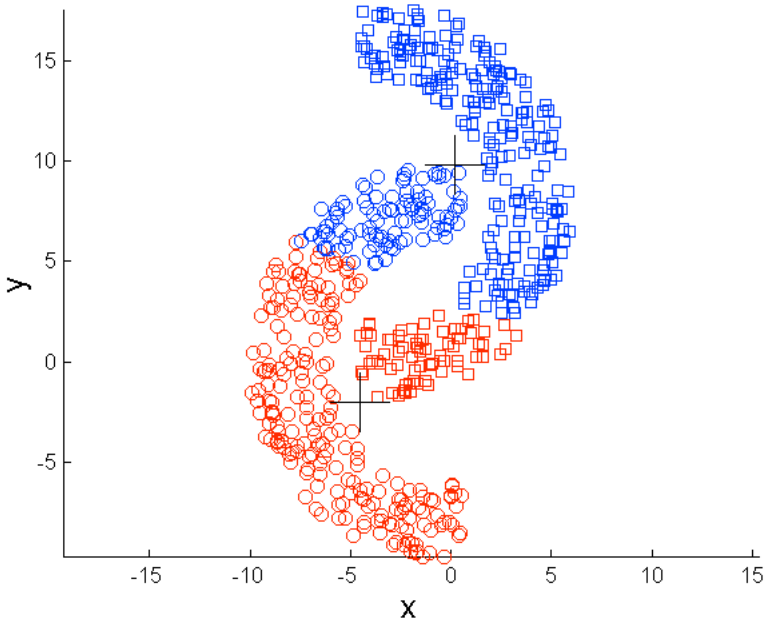
Limitations of K-means: Non-globular Shapes



Original Points



K-means (2 Clusters)

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From K-Means to K-Medoids

- The k-means is sensitive to outliers-since an object with an extremely large value may substantially distort the distribution of the data
- K-Medoids: Instead of taking the mean value of the object in a cluster as a reference point, medoids can be used, which is the most centrally located object in a cluster
- **K-Medoids**

1. Initialize: select k random points out of the n data points as the medoids.

2. Associate each data point to the closest medoid by using any common distance metric methods.

3. While the cost decreases:

For each medoid m, for each data o point which is not a medoid:

1. Swap m and o, associate each data point to the closest medoid, recompute the cost.

2. If the total cost is more than that in the previous step, undo the swap.

How to choose a “good” k for k-means clustering?

- Choose the number of clusters by visually inspecting data points
- **Elbow method:**
  - Compute the sum of squared error (SSE) for some values of k (for example, 2,3,‘ d as the sum of the squared distance between each member of the cluster and its centroid.

number of clusters

number of cases

case i

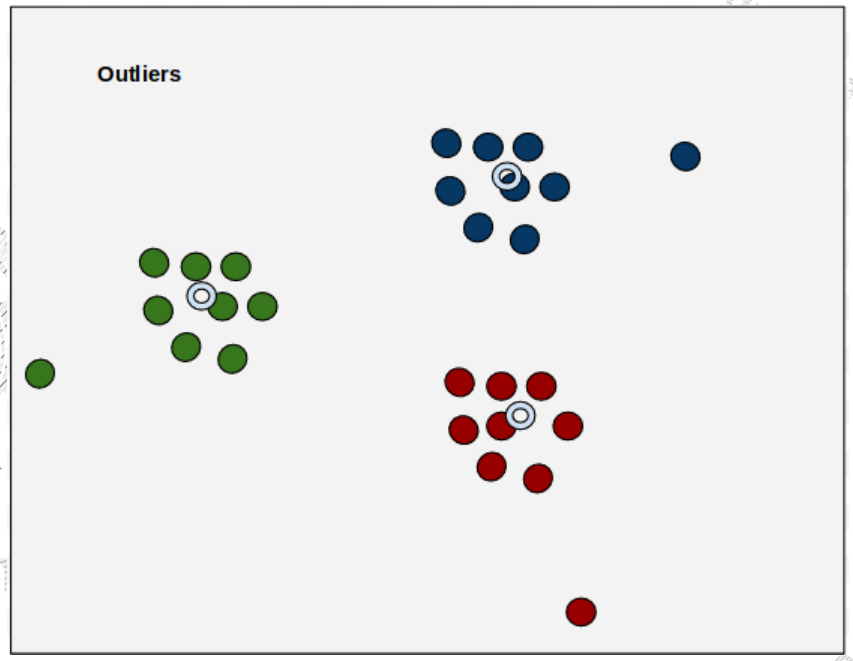
centroid for cluster j

objective function

$$J = \sum_{j=1}^{k} \sum_{i=1}^{n} \underbrace{\left\| x_i^{(j)} - c_j \right\|^2}_{\text{Distance function}}$$

(naive) K-Means for detecting outliers

- Let
  - dist(x, μ_k) be the distance of a point x, assigned to cluster k to its center μ_k.
  - Lμ_k be the the average distance of all the points assigned to cluster k with its center



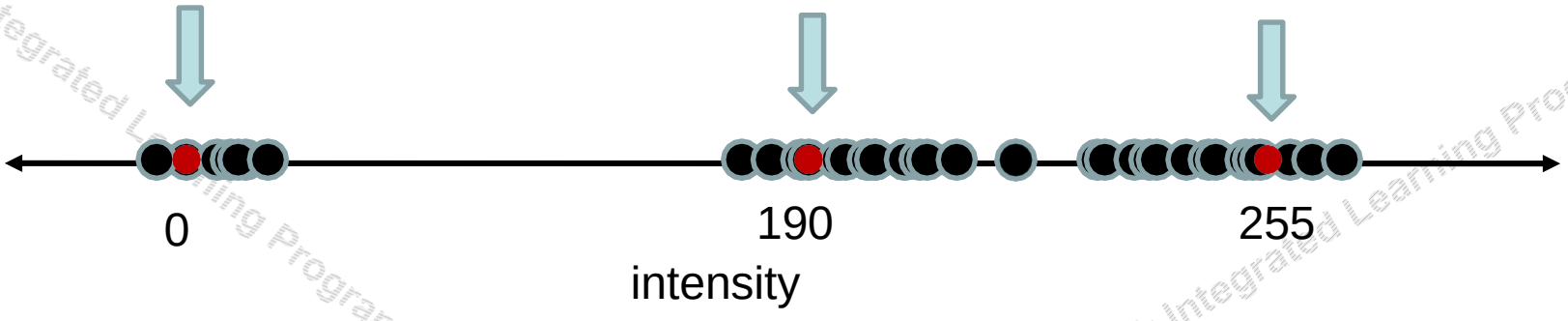
- The ratio dist(x, μ_k) / Lμ_k for each point is the outlier score for each point.
- Higher the ratio for a point x, more likely x is an outlier

Notion of Mixture Models

Work Integrated Learning Programmes

K means – Hard clustering

- Assigned each example to exactly one cluster
- What if clusters are overlapping?
  - Hard to tell which cluster is right
  - Maybe we should try to remain uncertain
- What if cluster has a non-circular shape?
- Hard assignments of data points to clusters – small shift of a data point can flip it to a different cluster
- For example, if a point is near the 'border' between two clusters, it's often better to know that it has near equal membership probabilities for these clusters, rather than blindly assigning it to the nearest one.



1 of K coding mechanism

- For each data point x_n, we introduce a set of binary indicator variables $r_{nk} \in [0,1]$ such that
- $$\sum_k r_{nk} = 1$$

where k =1,...,K describing which of the K clusters the data point x_n is assigned to, so that if data point x_n is assigned to cluster k then $r_{nk} > r_{nj}$ for j not equal to k.

- Example: 5 data points and 3 clusters

$$(r_{nk}) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}$$

GMM - soft Clustering

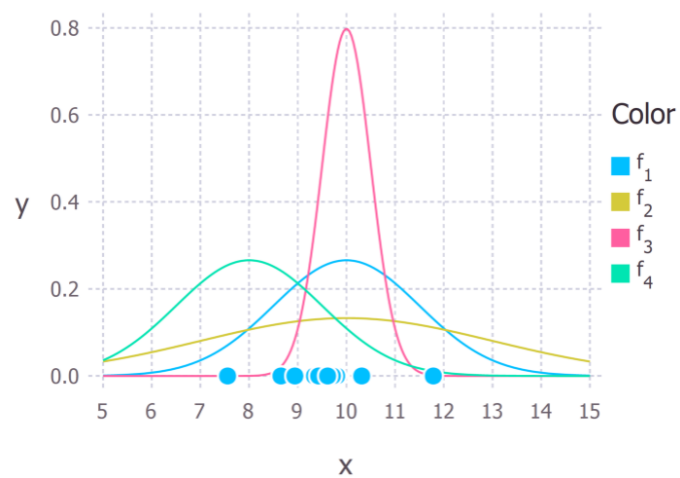
- **Solution**: replace ‘hard’ clustering of K-means with ‘soft’ probabilistic assignments
- Probabilistic Clustering
  - Represents the probability distribution of the data as a **Gaussian mixture model (GMM)**
  - GMMs give a probabilistic assignment of points to clusters which quantify uncertainty.
  - Mixture models combines a set of distributions to create a convex space where we can search for the optimal parameters for such distributions using MLE.
    - Mixture models in general don't require knowing which subpopulation a data point belongs to, allowing the model to learn the subpopulations automatically. Since subpopulation assignment is not known, this constitutes a form of unsupervised learning.
- In GMM, Clusters modeled as Gaussians
- **EM algorithm:** assign data to cluster with some probability

Source Credit: Christopher M. Bishop

Gaussian Distribution

- The normal curve is *bell-shaped* and has a single peak at the exact center of the distribution.
- The arithmetic mean, median, and mode of the distribution are equal and located at the peak.
- Half the area under the curve is above the peak, and the other half is below it.
- The normal distribution is symmetrical about its mean.

Holds	Fails
Amalgamation of similar distributions	Dominated by one (or few) particular distribution
Contributing factors are independent	Dependencies among contributing factors
Sample selection is uniformly random	Sample selection is correlated to previous selection



The 10 data points and possible Gaussian distributions from which the data were drawn. f1 is normally distributed with mean 10 and variance 2.25 (variance is equal to the square of the standard deviation), this is also denoted f1 ~ N (10, 2.25). f2 ~ N (10, 9), f3 ~ N (10, 0.25) and f4 ~ N (8, 2.25)

MATHEMATICAL FUNCTION (Pdf)

$$f(x) = \frac{1}{\boxed{\sigma}\sqrt{2\pi}} \cdot e^{-\frac{1}{2}\left(\frac{x-\boxed{\mu}}{\sigma}\right)^2}$$

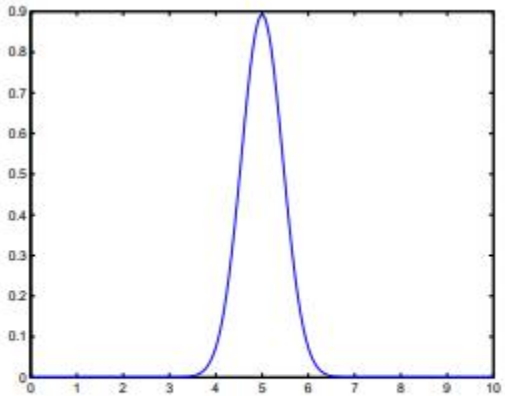
This is a bell shaped curve with different centers and spreads depending on μ and σ

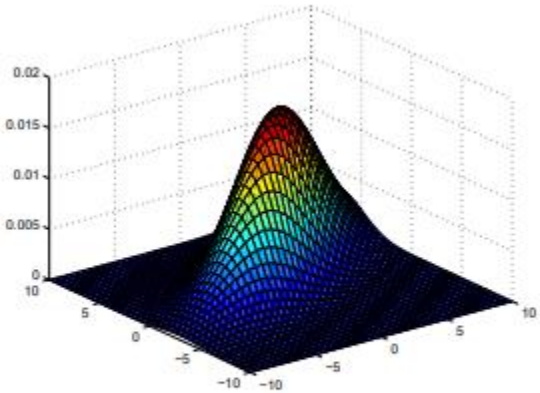
Note constants:

π=3.14159

e=2.71828

Univariate and Multivariate Gaussian density





If $\mathbf{x}$ is multivariate normal, it also has a **mean vector** μ such that

$$\mu = (\mathbb{E}(X_1), \mathbb{E}(X_2), \ldots, \mathbb{E}(X_n))$$

1.
2. where $\mathbb{E}(X)$ is the expected value (or mean) of X. $\mathbf{x}$ also has a **covariance matrix** Σ satisfying ample

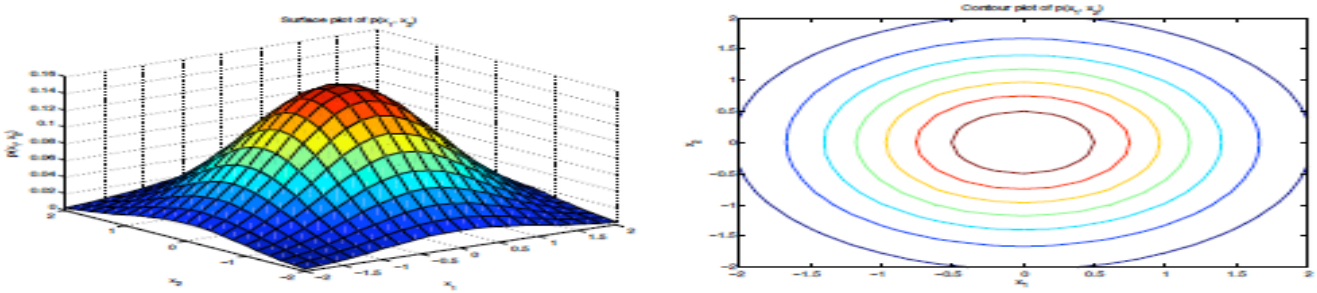
$$\Sigma_{i,j} = \mathrm{Cov}(X_i, X_j)$$

where $\mathrm{Cov}(X_i, X_j)$ is the covariance of X_i and X_j.

Furthermore, $\mathbf{x}$ is completely defined by μ and Σ, so it is convenient to write

$$\mathbf{x} \sim \mathcal{N}(\mu, \Sigma).$$

2-dimensional Gaussian distribution

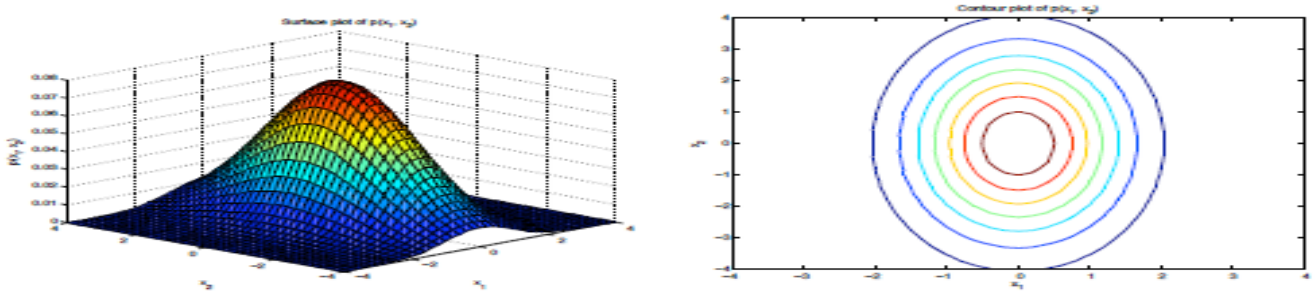


(a) Spherical Gaussian (diagonal covariance, equal variances)

A diagonal covariance matrix means

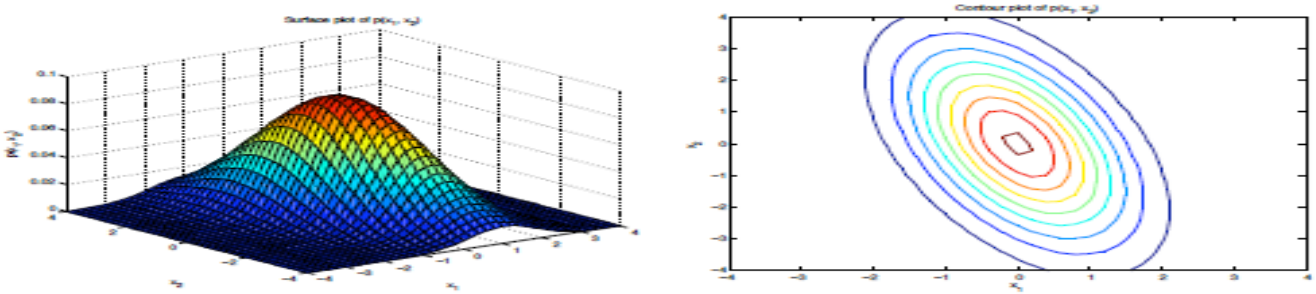
that the variables are uncorrelated,

and the matrix only has values on its diagonal.



(b) Gaussian with diagonal covariance matrix

Unequal variances: there is still no correlation between the two dimensions



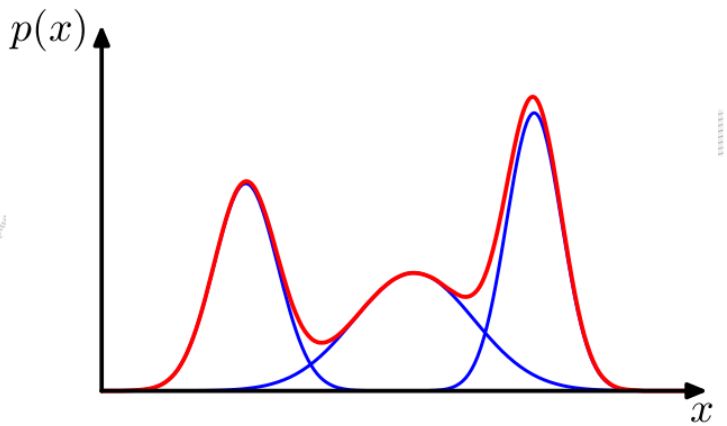
(c) Gaussian with full covariance matrix

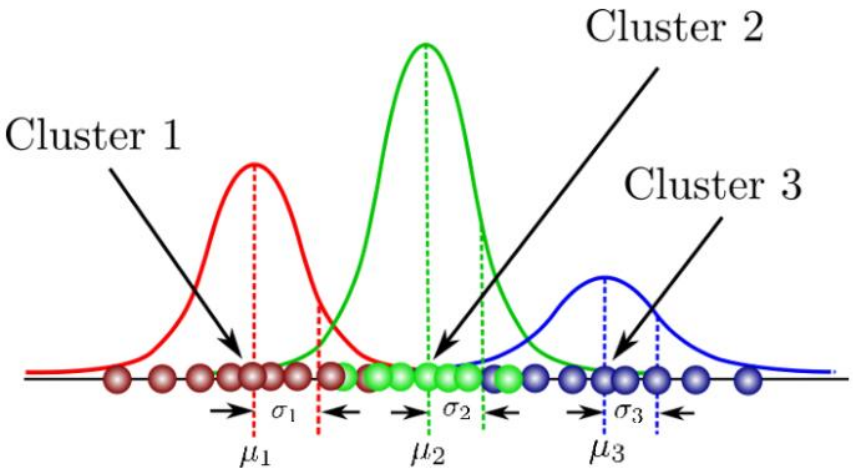
Full covariance: correlation between teh dimensions

Figure 7: Surface and contour plots of 2–dimensional Gaussian with different covariance structures

GMM – Mixture of Gaussians

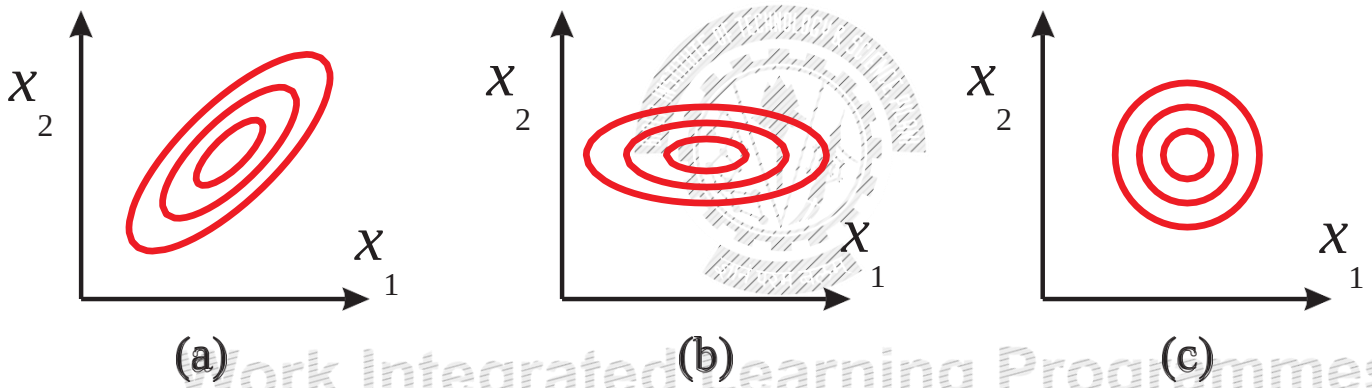
- Probabilistic cluster is a distribution over the data space, which can be mathematically represented using a probability density function (or distribution function).
- Our task: infer a set of k probabilistic clusters that is mostly likely to generate D





Example: Mixture of 3 Gaussians

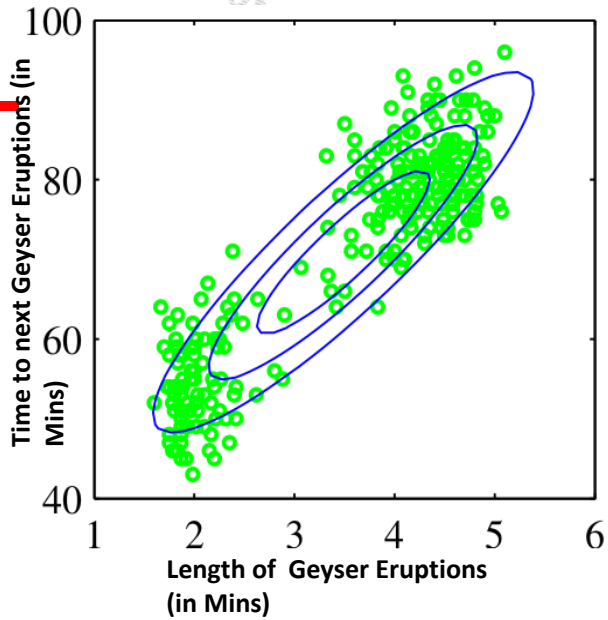
$$\mathcal{N}(\mathbf{x}|\boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{d/2}|\boldsymbol{\Sigma}|^{1/2}} \exp\left\{-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^{\top}\boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})\right\}$$

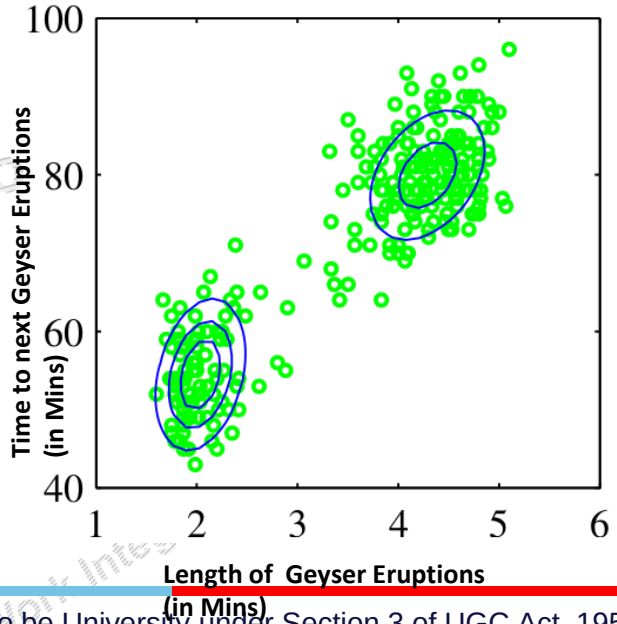


- The observed data come from a Gaussian mixture distribution which consists of K Gaussians with their own means and variances.
- K classes are latent

GMM – Why single distribution is insufficient

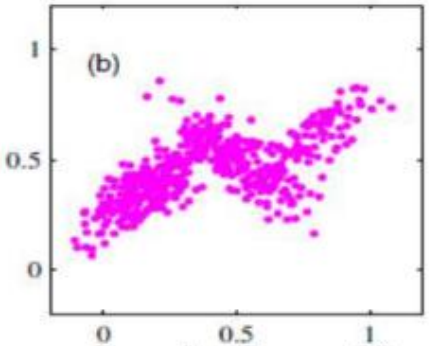
- Gaussian distribution is that it is intrinsically unimodal
- A single distribution is not sufficient to fit many real data sets which is multimodal
- A mixture model assumes that a set of observed objects is a mixture of instances from multiple probabilistic clusters, and conceptually each observed object is generated independently





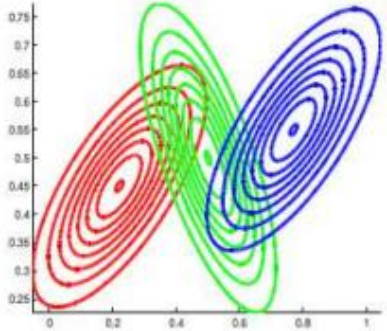
Learning a Mixture of Gaussians

Our actual observations

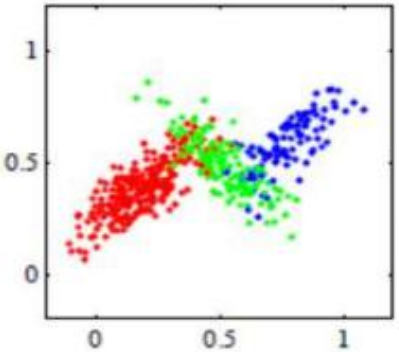


Samples from p(x)

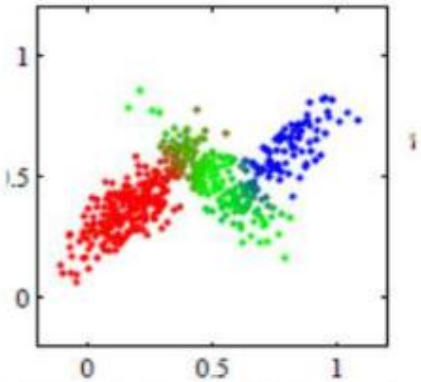
Mixture of 3 Gaussians



Component Distributions



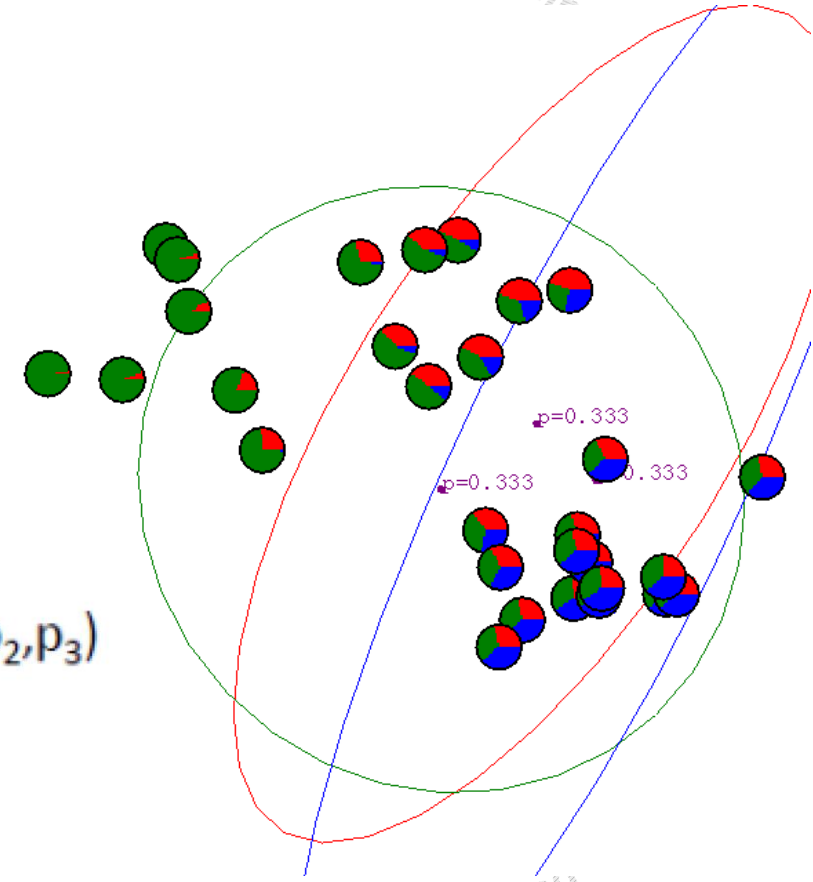
Samples labeled using their true component



Soft clustering learned by a Gaussian mixture model

Source Credit :CS229: Machine Learning ©2021 Carlos Guestrin

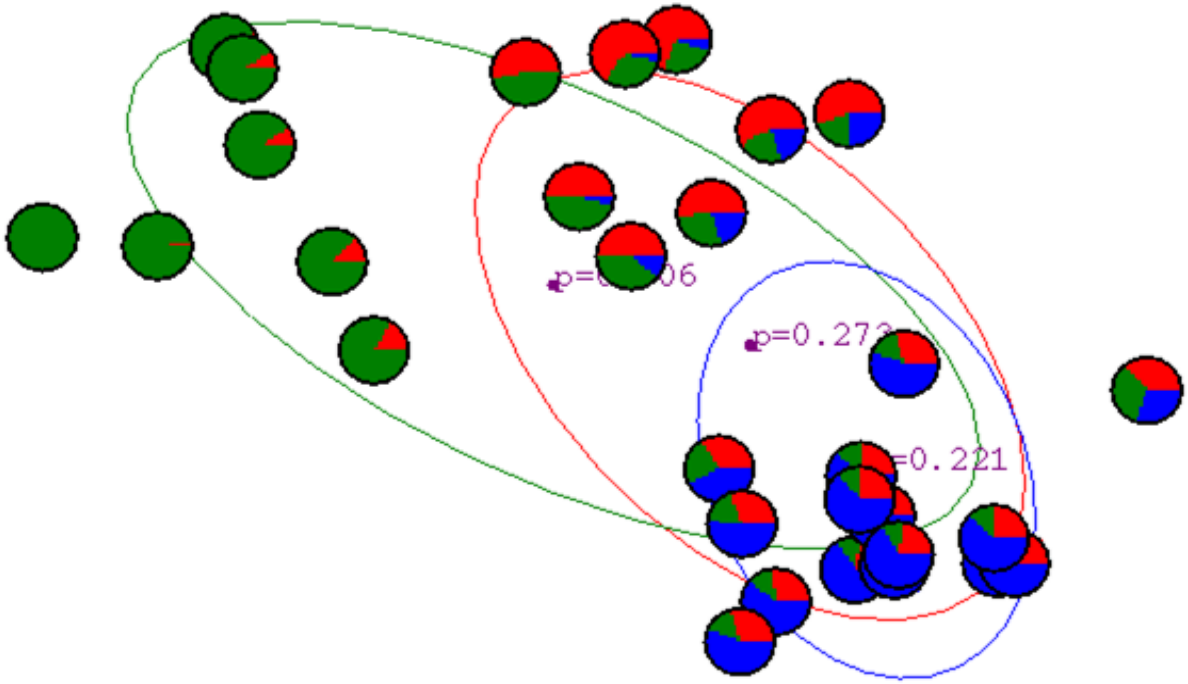
Gaussian Mixture Example: Start



$$P(y = \bullet \mid x_j, \mu_1, \mu_2, \mu_3, \Sigma_1, \Sigma_2, \Sigma_3, p_1, p_2, p_3)$$

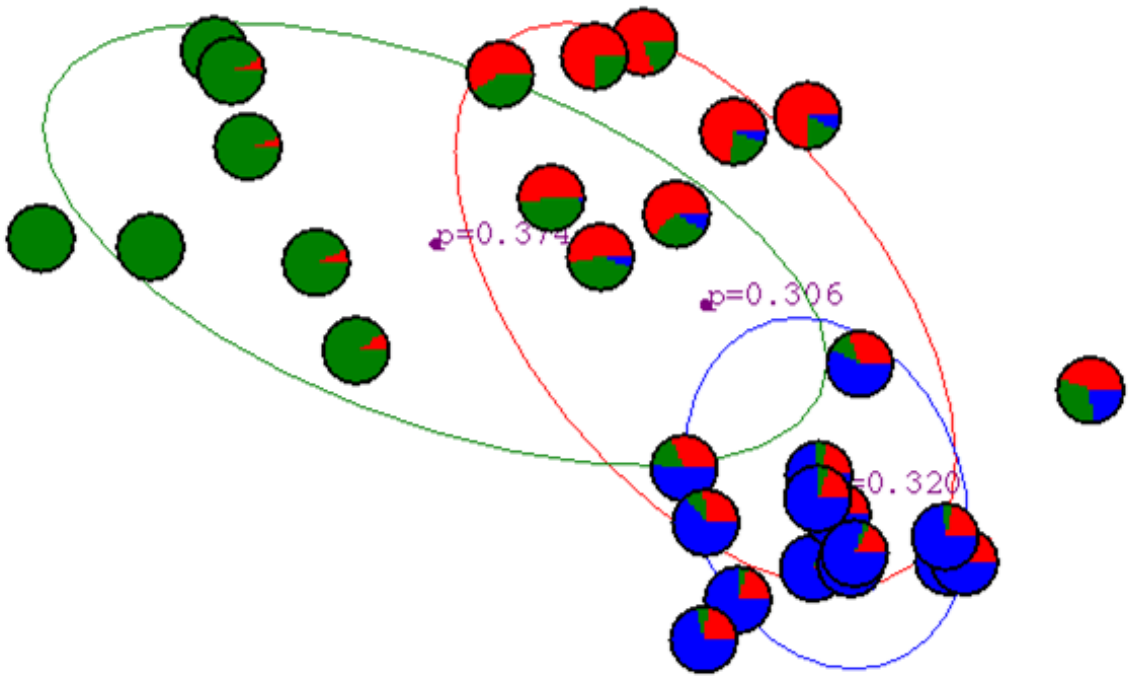
Slide Credit: Andrew W. Moore

After first iteration



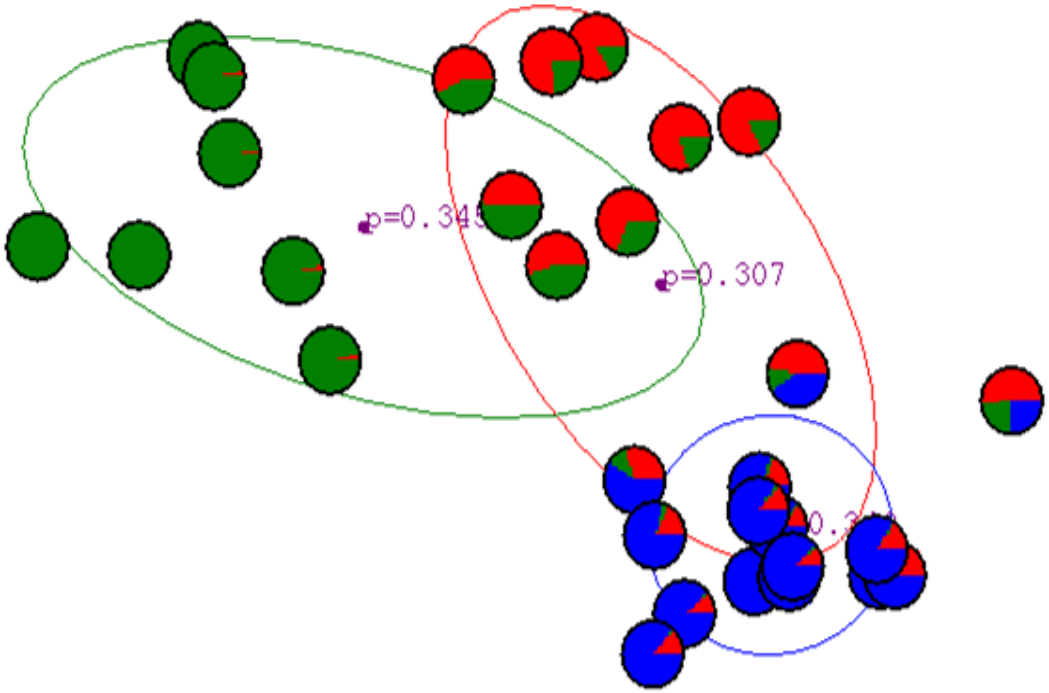
Slide Credit: Andrew W. Moore

After 2nd iteration



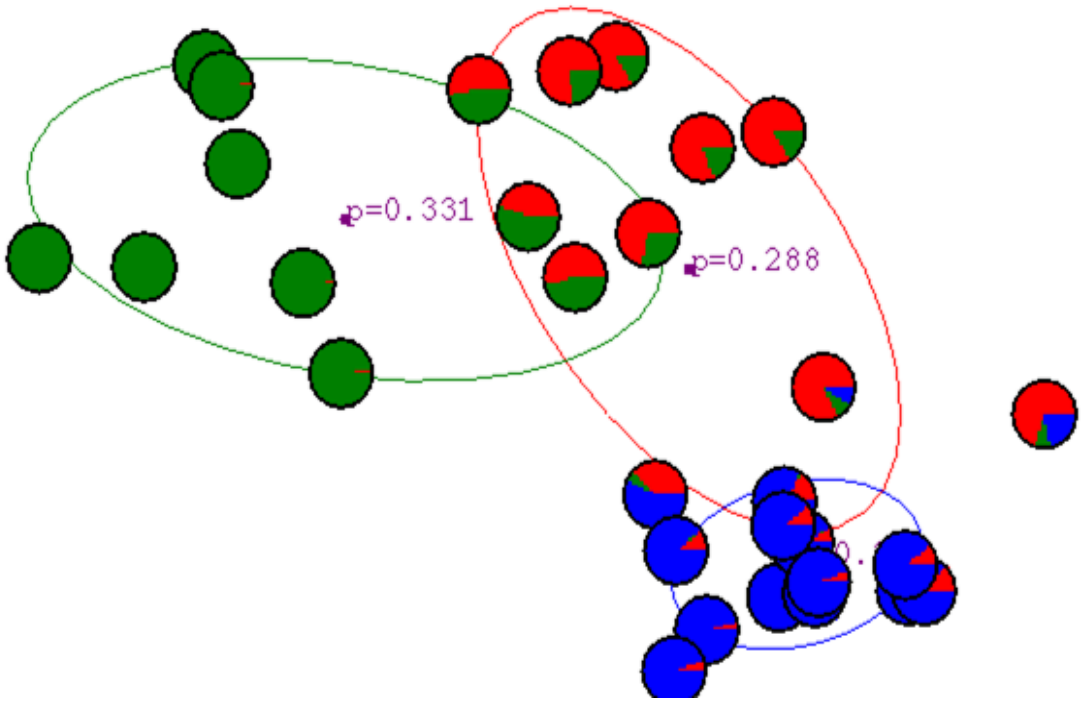
Slide Credit: Andrew W. Moore

After 3rd iteration



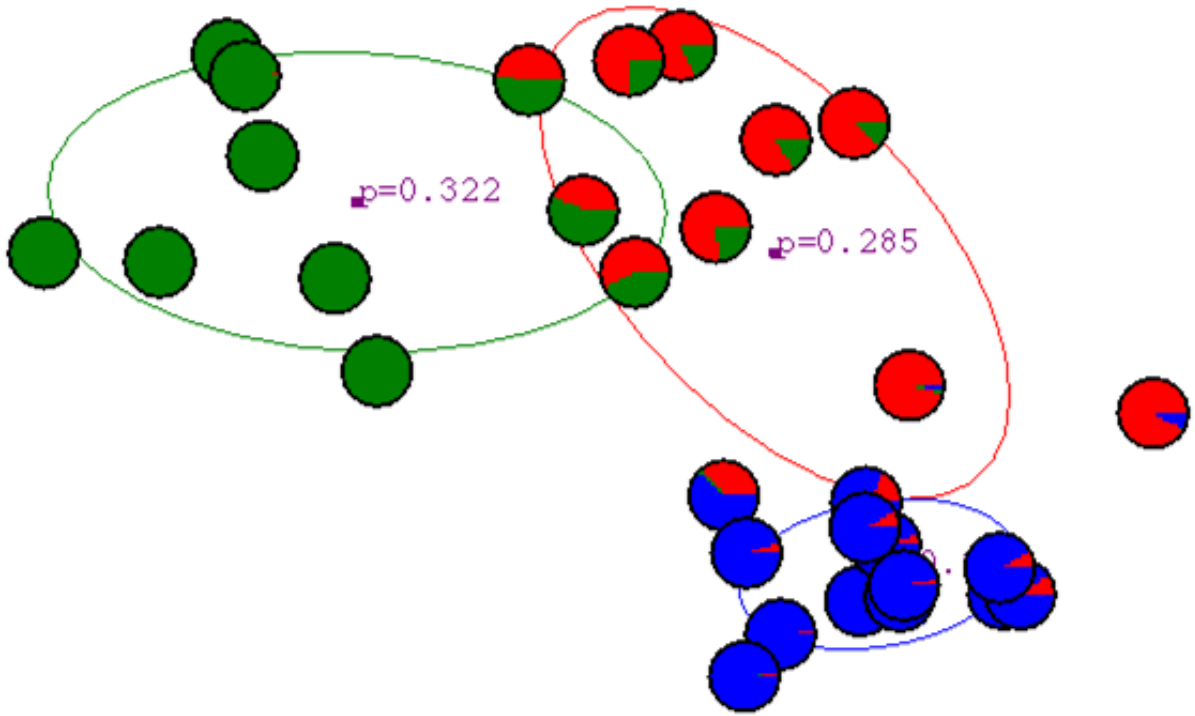
Slide Credit: Andrew W. Moore

After 4th iteration



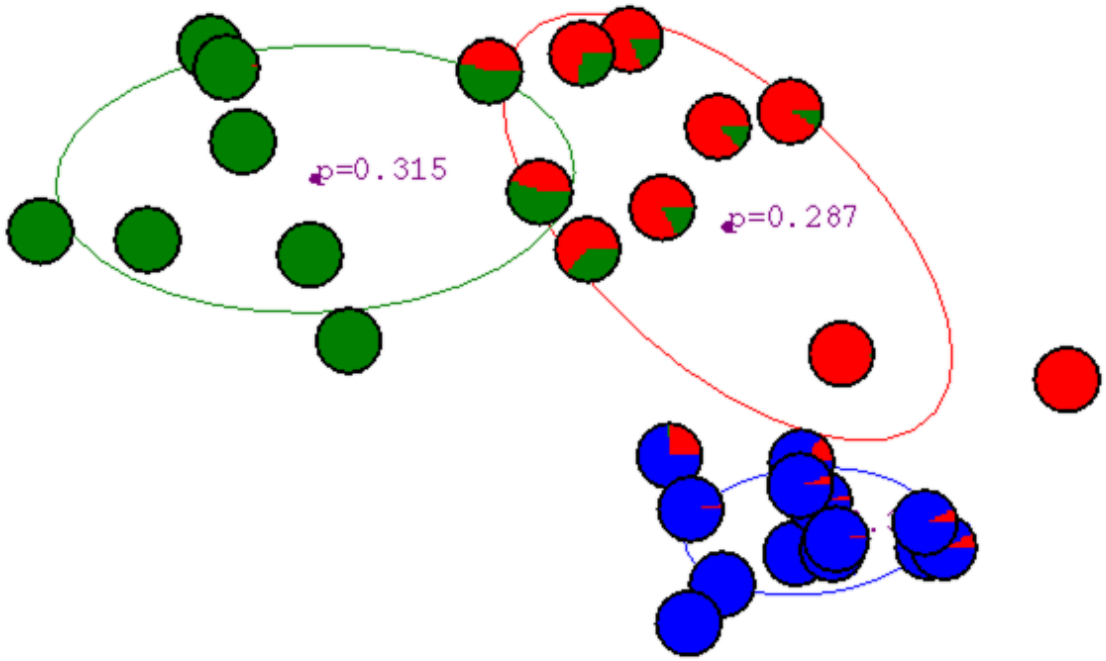
Slide Credit: Andrew W. Moore

After 5th iteration



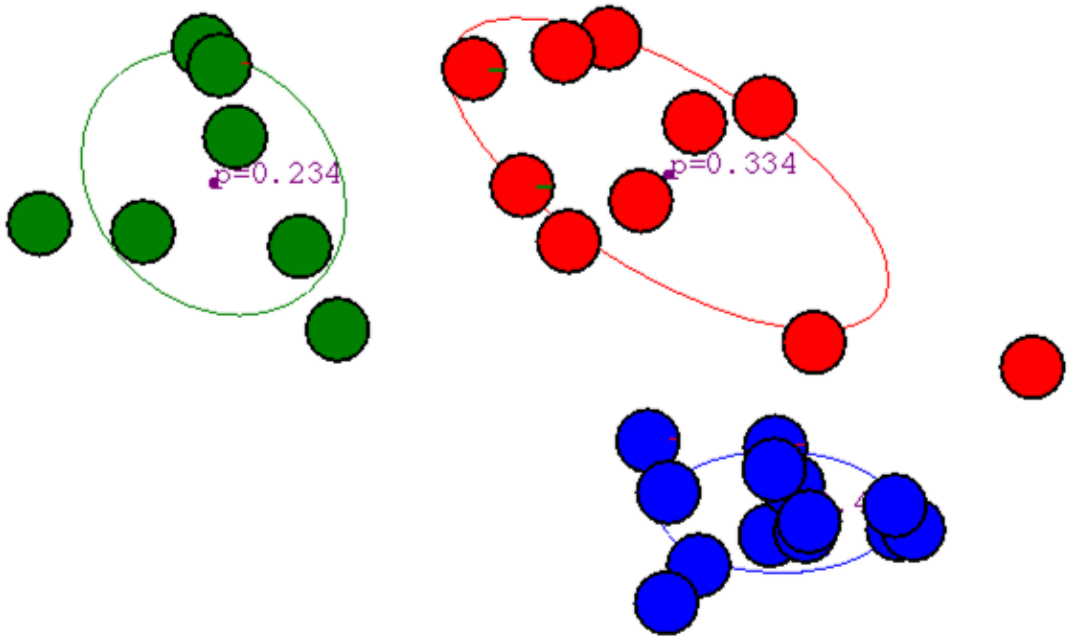
Slide Credit: Andrew W. Moore

After 6th iteration



Slide Credit: Andrew W. Moore

After 20th iteration



Slide Credit: Andrew W. Moore

GMM equations

$$p(\mathbf{x}) = \sum_{\mathbf{z}} p(\mathbf{z})p(\mathbf{x}|\mathbf{z})$$

Mixture models

K = number of mixture components (clusters),

π_j = mixture weights

$p_j(x)$ = family of distributions

(Gaussian, Bernoulli, etc).

$$p(\mathbf{x}) = \sum_{j=1}^{K} \pi_j p_j(\mathbf{x})$$

$$0 \leq \pi_j \leq 1, \qquad \sum_{j=1}^{K} \pi_j = 1$$

Gaussian Mixture models

$$p(\mathbf{x}|\boldsymbol{\theta}) = \sum_{j=1}^{K} \pi_j \mathcal{N}(\mathbf{x}|\boldsymbol{\mu_j}, \boldsymbol{\Sigma_j})$$

$$0 \leq \pi_j \leq 1, \qquad \sum_{j=1}^{K} \pi_j = 1$$

θ = collection of all the parameters of the model (mixture weights, means, and covariance matrices):

$$\boldsymbol{\theta} := \{\pi_1, \cdots, \pi_K, \boldsymbol{\mu_1}, \cdots, \boldsymbol{\mu_K}, \boldsymbol{\Sigma_1}, \cdots, \boldsymbol{\Sigma_K}\}$$

General EM algorithm

Observed data: D={x1, …….., xn}

Unknown variables: y

In clustering: y=1….K clusters

Parameters: θ

In GMM : θ= π : {π1, . . . , πk}

μ : {μ1 , . . . , μK}

Σ : {Σ1, . . . ΣK }

Goal: $\hat{\theta}_n = \arg\max_{\theta} \log P(D|\theta)$

In GMM : $$\ln p(\mathbf{X}|\boldsymbol{\pi}, \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \sum_{n=1}^{N} \ln \left\{ \sum_{k=1}^{K} \pi_k \mathcal{N}(\mathbf{x}_n|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k) \right\}$$

EM algorithm for GMM

E step:

Evaluate the responsibilities (posterior probabilities) using the current parameter

$$\gamma(z_k) \equiv p(z_k = 1|\mathbf{x}) = \frac{p(z_k = 1)p(\mathbf{x}|z_k = 1)}{\sum_{j=1}^{K} p(z_j = 1)p(\mathbf{x}|z_j = 1)} = \frac{\pi_k \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_{j=1}^{K} \pi_j \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)}.$$

- For each example x_n,
- Compute $\gamma(z_{nk})$ the probability that x_n is generated by component Z_k i.e it belongs to cluster k
- If x_n is very likely under the k_{th} Gaussian, it gets high weight

EM algorithm for GMM

M-Step :

- maximize the expectation of the complete-data log-likelihood, computed with respect to the conditional probabilities found in the Expectation step. The result of the maximization is a new parameter vector μnew , Σnew and πnew
- Keep γ (znk) fixed, and apply MLE for maximizing of ln p(X|π,μ,Σ) for μk , Σk and πk, to get μnew , Σnew and πnew

$$\ln p(\mathbf{X}|\boldsymbol{\pi}, \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \sum_{n=1}^{N} \ln \left\{ \sum_{k=1}^{K} \pi_k \mathcal{N}(\mathbf{x}_n | \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k) \right\}$$

Repeat E & M until convergence

- In practice, the algorithm is deemed to have converged when the change in the log likelihood function, or alternatively in the parameters, falls below some threshold

EM algorithm for GMM

To Estimate:

M-Step

Initialize $\boldsymbol{\pi}$, μ,Σ
and also evaluate the log likelihood

E-Step

$$\gamma(z_k) \equiv p(z_k = 1|\mathbf{x}) = \frac{p(z_k = 1)p(\mathbf{x}|z_k = 1)}{\sum_{j=1}^{K} p(z_j = 1)p(\mathbf{x}|z_j = 1)} = \frac{\pi_k \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_{j=1}^{K} \pi_j \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)}.$$

Perform E-Step Given $\gamma(z_k)$

$$\boldsymbol{\mu}_k^{\text{new}} = \frac{1}{N_k}\sum_{n=1}^{N}\gamma(z_{nk})\mathbf{x}_n$$

$$\boldsymbol{\Sigma}_k^{\text{new}} = \frac{1}{N_k}\sum_{n=1}^{N}\gamma(z_{nk})\left(\mathbf{x}_n - \boldsymbol{\mu}_k^{\text{new}}\right)\left(\mathbf{x}_n - \boldsymbol{\mu}_k^{\text{new}}\right)^{\mathrm{T}}$$

$$\pi_k^{\text{new}} = \frac{N_k}{N}$$

$$N_k = \sum_{n=1}^{N}\gamma(z_{nk})$$

Perform M-Step Given $\boldsymbol{\pi}$, μ,Σ

$$p(z_k = 1) = \pi_k$$

Repeat Until
Convergence

N_k = the effective number of points assigned to cluster k

EM Algorithm

	Word1	Word2
Doc1	2	2
Doc2	4	5
Doc3	7	2

	Means	SDs	Weight
Cluster 1	(3,3)	(4,0.707)	0.5
Cluster 2	(4,4)	(0.5,0.707)	0.5

I Standardize the data if required:

II Fix the no.of.cluster expected

III Initialize the prototypes:
Mean, Covariance, Weights

IV Expectation-Step: Fix prototype & find the membership of each point weighted by the probability value

V. Calculate the log likelihood of the points

VI Maximization Step: Fix the membership(responsibility matrix) and re-estimate the prototypes

VII Calculate the new log likelihood of the points. Repeat E & M Step till convergence is achieved:

Work Integrated Learning Programmes

GMM

	Word1	Word2
Doc1	2	2
Doc2	4	5
Doc3	7	2

	Means	SDs	Weight
Cluster 1	(3,3)	(4,0.707)	0.5
Cluster 2	(4,4)	(0.5,0.707)	0.5

IV Expectation-Step: Fix prototype & find the membership of each point weighted by the probability value

$$\gamma(z_k) \equiv p(z_k = 1|\mathbf{x}) = \frac{\pi_k \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_{j=1}^{K} \pi_j \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)}.$$

<u>Sample calculation for this step is shown below:</u>

	Doc1	Doc2	Doc3
Weight1 * N1			
Weight2 * N2			
Sum			
P(Zi1)			
P(Zi2)			

Weight1*N1 =

0.5* $P(Doc1, Cluter1)$

$= (Doc1, (3,3), (4,0.707))$

$$= 0.5 * \frac{e^{\frac{-(2-3)^2}{2*4^2}}}{4*\sqrt[2]{2*3.14}} * \frac{e^{\frac{-(2-3)^2}{2*0.707^2}}}{0.707*\sqrt[2]{2*3.14}}$$

$= 0.01004$

For Cluster 1 assumed parameters, here for Doc1 apply in the below formula and multiply with the weight = 0.5. For simplicity , assume off diagonal values in covariance matrix as 0 (This is usually done in text analysis use cases). similarly do it for all the Doc's

$$p(x; \mu, \Sigma) = \frac{1}{(2\pi)^{n/2}|\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)\right)$$

GMM

	Word1	Word2
Doc1	2	2
Doc2	4	5
Doc3	7	2

	Means	SDs	Weight
Cluster 1	(3,3)	(4,0.707)	0.5
Cluster 2	(4,4)	(0.5,0.707)	0.5

IV Expectation-Step: Fix prototype & find the membership of each point weighted by the probability value

$$\gamma(z_k) \equiv p(z_k = 1|\mathbf{x}) = \frac{\pi_k \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_{j=1}^{K} \pi_j \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)}.$$

	Doc1	Doc2	Doc3
Weight1 * N1			
Weight2 * N2			
Sum			
P(Zi1)			
P(Zi2)			

Similarly:

For Cluster 2 assumed parameters, here for Doc1 apply in the below formula and multiply with the weight = 0.5. For simplicity , assume off diagonal values in covariance matrix as 0(This is usually done in text analysis use cases). similarly do it for all the Doc's

$$p(x; \mu, \Sigma) = \frac{1}{(2\pi)^{n/2}|\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)\right)$$

GMM

	Word1	Word2
Doc1	2	2
Doc2	4	5
Doc3	7	2

	Means	SDs	Weight
Cluster 1	(3,3)	(4,0.707)	0.5
Cluster 2	(4,4)	(0.5,0.707)	0.5

$$\gamma(z_k) \equiv p(z_k = 1|\mathbf{x}) = \frac{\pi_k \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_{j=1}^{K} \pi_j \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)}$$

	Doc1	Doc2	Doc3
Weight1 * N1			
Weight2 * N2			
Sum			
P(Zi1)			
P(Zi2)			

Per Docs sum the previous two found values.
This gives the denominator of this formula

For Doc2 :
Weight1*N1 = 0.0005
Weight2*N2 = 0.0828
Sum = 0.08329 = ~0.0833

$P(Z_{Doc1\text{-}Cluster1}) = 0.006$
$P(Z_{Doc1\text{-}Cluster1}) = 0.994$

Now find the membership of doc :
P(Zi1) = dividing Weight1*N1 by Sum
P(Zi2) = dividing Weight2*N1 by Sum

Similarly do it for all the Doc's

GMM

	Word1	Word2
Doc1	2	2
Doc2	4	5
Doc3	7	2

	Means	SDs	Weight
Cluster 1	(3,3)	(4,0.707)	0.5
Cluster 2	(4,4)	(0.5,0.707)	0.5

V. Calculate the log likelihood of the points

$$\gamma(z_k) \equiv p(z_k = 1|\mathbf{x}) = \frac{\pi_k \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_{j=1}^{K} \pi_j \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)}.$$

	Doc1	Doc2	Doc3
Weight1 * N1			
Weight2 * N2			
Sum			
P(Zi1)			
P(Zi2)			

Log likelihood = ln(sum of Doc1) + ln(sum of Doc2 + ln(Sum of Doc3)....

Log likehood =
ln(0.010) + ln(0.083) + ln(0.006)
=-12.16

This value needs to be found for every iteration and at one iteration it will approach a very large value . That will be one of the stopping convergence point of the algorithm.

GMM

	Word1	Word2
Doc1	2	2
Doc2	4	5
Doc3	7	2

	Means	SDs	Weight
Cluster 1	(3,3)	(4,0.707)	0.5
Cluster 2	(4,4)	(0.5,0.707)	0.5

VI Maximization Step: Fix the membership(responsibility matrix) and re-estimate the prototypes

	Doc1	Doc2	Doc3
P(Zi1)	0.9999	0.006	1
P(Zi2)	0.0001	0.994	0

	Means	SDs	Weight
Cluster 1			
Cluster 2			

<u>Sample calculations shown below</u>

$$N_k = \sum_{n=1}^{N} \gamma(z_{nk})$$

N1 for cluster k=1 is got by summing these. Similarly do it for N2

$N_{k=1}$ = Cluster 1 =
= 0.9999+0.006+1
=2.0059
This is the value N1 represented here

For cluster k=1 is Means obtained by multiplying the data by the membership.

[0.9999(2) + 0.006(4) + 1(7)]/N1 , [0.9999(2) + 0.006(5) + 1(2)]/N1

$MEAN_{k=1}$ =
(4.49, 2.00) = ~(4.49,2)

$$\boldsymbol{\mu}_k^{\text{new}} = \frac{1}{N_k} \sum_{n=1}^{N} \gamma(z_{nk}) \mathbf{x}_n$$

Similarly do it for cluster 2

$$\boldsymbol{\Sigma}_k^{\text{new}} = \frac{1}{N_k} \sum_{n=1}^{N} \gamma(z_{nk}) (\mathbf{x}_n - \boldsymbol{\mu}_k^{\text{new}}) (\mathbf{x}_n - \boldsymbol{\mu}_k^{\text{new}})^{\mathrm{T}}$$

$Std.Dev_{k=1}$ =
([0.9999(2−4.49)^2+*0.006*(4−4.49)^2+*1*(7−4.49)^2]/*N1* ,
[0.9999(2−2)^2+*0.006*(5−2)^2+*1*(2−2)^2]/*N1*)

= (6.23, 0.03)

$$\pi_k^{\text{new}} = \frac{N_k}{N}$$

$Weight_{k=cluster1\ (new)}$
= 2.0059/3 = 0.67

GMM - EM Algorithm

2nd Iteration continues

	Word1	Word2
Doc1	2	2
Doc2	4	5
Doc3	7	2

I Standardize the data if required:

II Fix the no.of.cluster expected

III Initialize the prototypes: Mean, Covariance, Weights

IV Expectation-Step: Fix prototype & find the membership of each point weighted by the probability value

V. Calculate the log likelihood of the points

VI Maximization Step: Fix the membership(responsibility matrix) and re-estimate the prototypes

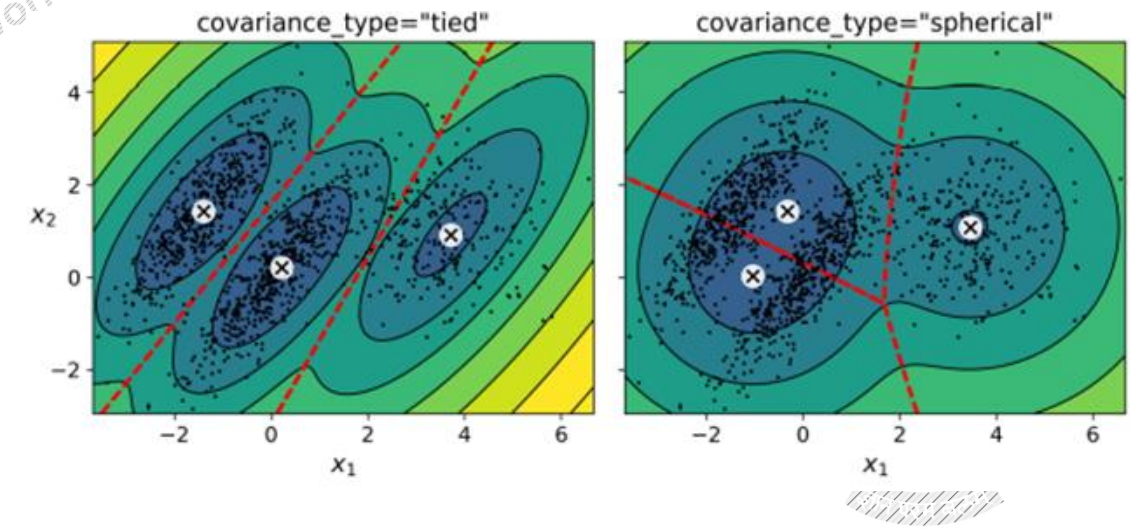
VII Calculate the new log likelihood of the points. Repeat E & M Step till convergence is achieved.

$$\gamma(z_k) \equiv p(z_k = 1|\mathbf{x}) = \frac{\pi_k \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_{j=1}^{K} \pi_j \mathcal{N}(\mathbf{x}|\boldsymbol{\mu}_j, \boldsymbol{\Sigma}_j)}.$$

	Means	SDs	Weight
Cluster 1	(4.49,2)	(6.32,0.03)	0.67
Cluster 2	(3.99,4.99)	(0.001,0.001)	0.33

	Doc1	Doc2	Doc3
Weight1 * N1			
Weight2 * N2			
Sum			
P(Zi1)			
P(Zi2)			

Gaussian Mixture Model



Note:
The GMM model can be relaxed by setting these parameters while calling the function

- "**spherical**": all clusters must be spherical, but they can have different diameters (i.e., different variances).
- "**diag**": clusters can take on any ellipsoidal shape of any size, but the ellipsoid's axes must be parallel to the coordinate axes (i.e., the covariance matrices must be diagonal).
- "**tied**": all clusters must have the same ellipsoidal shape, size and orientation (i.e., all clusters share the same covariance matrix).

Gaussian Mixture Model

```
from sklearn.mixture import GaussianMixture

gm = GaussianMixture(n_components=3, n_init=10)
gm.fit(X)

gm.weights_
gm.means_
gm.covariances_
```

```
>>> gm.predict(X)
array([2, 2, 1, ..., 0, 0, 0])
>>> gm.predict_proba(X)
array([[2.32389467e-02, 6.77397850e-07, 9.76760376e-01],
       [1.64685609e-02, 6.75361303e-04, 9.82856078e-01],
```